

SE YOON LEE

1002 Neubauer Cir, IL, 60046, U.S.A.

(979) · 888 · 4107 ◇ seyoonlee.stat.math@gmail.com

<https://sites.google.com/view/seyoonlee>

PROFESSIONAL SUMMARY

- An experienced statistician with 5 years of experience in the healthcare industry, including permanent employment and internships. Extensive experience in study designs, statistical operations, and scientific publications across a broad range of therapeutic areas, including cardiovascular disease, oncology, and ophthalmology
- Major functional contributor to regulatory approvals in numerous product and clinical applications across the globe. Specializing in Bayesian adaptive design, complex modeling and simulation, data integration, systematic reviews and meta-analysis, and scientific publications
- Expert in R/SAS for the development of simulation-based sample size calculations for complex and innovative trial designs and NONMEM for population pharmacokinetic/pharmacodynamic (PK/PD) models
- Author of many top peer-reviewed statistical and scientific articles, and invited peer-reviewer for top-tier statistical methodology journals and scientific journals.

EDUCATION

Ph.D. Statistics, Texas A&M University, College Station, U.S.A.

Advisor: Prof. Bani K. Mallick

May 2021

Overall GPA: 3.78/4

M.A. Applied Statistics, Yonsei University, Seoul, South Korea

Advisor: Prof. Joseph H.T. Kim

Feb 2016

Overall GPA: 3.86/4

B.S. Mathematics, Yonsei University, Seoul, South Korea

Graduated with the 2nd highest class rank out of 43 students

Feb 2013

Overall GPA: 3.84/4

RESEARCH INTEREST & SPECIALTY

Clinical Trials, Bayesian Adaptive Design, Group Sequential Design, Survival Analysis, Modeling & Simulation

DISSERTATION AND THESIS

- “Bayesian Hierarchical Modeling: Application towards Complex and High-dimensional Data,” *Texas A&M University*, Doctoral Dissertation ; [\[Link\]](#)
- “Exponentiated generalized Pareto distribution: an alternative to the generalized Pareto distribution,” *Yonsei University*, Master’s Thesis ; [\[Link\]](#)

WORK EXPERIENCE

AbbVie.

Manager, Statistics (Permanent Employment)

Feb 2025 – present

North Chicago, IL

- Specialized in real-world data analysis and causal inference methods for regulatory and observational research.
- Specialized in developing simulation-based R programs for power and sample size calculation in Phase II and Phase III oncology trial designs at the regulatory submission level.
- Researched modern causal inference methods, including targeted maximum likelihood estimation, doubly robust estimation, and external control methodologies for clinical trial applications.
- Presented methodological research at statistical conferences, including JSM, on topics such as survival extrapolation, multiple imputation, and innovative clinical trial design.

Edwards Lifesciences.*May 2023 – Jan 2025***Manager, Biostatistics (Permanent Employment)**

Irvine, CA

- Developed an R code to perform unblinded sample size re-estimation (promising zone approach) using CHW test statistics for a hierarchical composite endpoint, employing the Finkelstein and Schoenfeld methodology.
- Specialized in developing simulation R code for power calculation of Bayesian adaptive design for cardiovascular trials at the regulatory submission level, with expertise in end-to-end coding and Statistical Analysis Plan writing.
- Provided seminars on the topic of Bayesian Medical Device Trials to train biostatisticians ; [\[Link\]](#)
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to cross-functional teams.

Johnson & Johnson.*Apr 2022 – May 2023***Senior Biostatistician (Permanent Employment)**

Irvine, CA

- Developed Bayesian adaptive clinical trials designs to evaluate the safety and effectiveness of medical devices (Bayesian basket trial designs, power-prior designs, Bayesian group sequential designs, etc).
- Provided advanced biostatistical supports (e.g., power analysis, primary endpoint analyses, clinical trial designs, etc) to clinicians with main focus on cardiovascular diseases.
- Led working groups of experts (Machine Learning/Statistical Methodology & Consulting) as leader
- Led the development of basket trial designs and authored Statistical Analysis Plan for SECURE, a Post-Market Study [\[ClinicalTrials.gov ID: NCT04750798\]](#)

Amgen Inc.*May 2021 – Apr 2022***Scientist - Modeling & Simulations (Permanent Employment)**

Thousand Oaks, CA

- Developed Pharmacokinetics/Pharmacodynamics/Cox hazard regression models for Phase I cancer clinical trials of AMG160 to treat a metastatic castration-resistant prostate cancer (mCRPC).
- Researched machine learning and deep neural network models to fit single dose data for subsequent simulation of multiple dosing scenarios.

Novartis International AG.*May 2020 – Aug 2020***Biostatistics Summer Intern (Contract Employment)**

East Hanover, NJ

- Developed a Bayesian linear mixed effect model for patients who have wet age-related macular degeneration to predict best-corrected visual acuity over the maintenance phase and suggest a personalized dose regimen. The proposed model has been trained by actual patients' data from HAWK and HARRIER studies (Number of patients is around 1,800 patients). ; [\[Abstract\]](#)

EMD Serono Inc. Merck KGaA.*May 2019 – Aug 2019***Pharmacometrics Summer Intern (Contract Employment)**

Billerica, MA

- Developed a Bayesian adaptive clinical trial design in Phase I cancer clinical trials, which aimed at utilizing grade information from the Common Toxicity Criteria for Adverse Events provided by the National Cancer Institute. ; [\[Abstract\]](#) ; [\[Poster\]](#)

Texas A&M University.*Aug 2016 – May 2021***Graduate Research/Teaching Assistant**

College Station, TX

- Researched on applications and developments of various statistical models (e.g. non-linear mixed effect model, nonparametric/semiparametric longitudinal model, clustering analysis, classification, hierarchical Poisson model, etc) to various industrial problems arising from biomedical, petroleum, wind energy industries, and COVID-19 outbreak.

PUBLICATIONS

Journal Article

- [1] **Seyoon Lee**, Joseph H.T. Kim. (2018) “Exponentiated generalized Pareto distribution: Properties and applications towards extreme value theory,” *Communications in Statistics - Theory and Methods*, 48:8, 2014-2038
- [2] **Se Yoon Lee**, Bowen Lei, and Bani K. Mallick. (2020) “Estimation of COVID-19 spread curves integrating global data and borrowing information,” *PLOS ONE* ; [[Github](#)]
- [3] **Se Yoon Lee**^{*}, Kahkashan Afrin^{*}, Ashif Iquebal^{*}, Mostafa Karimi^{*}, Allyson Larsen^{*}, and Bani K Mallick^{*}. (2020) “Directionally Dependent Multi-View Clustering Using Copula Model,” *PLOS ONE* (^{*} : equal contribution, authors are alphabetically ordered in the last name.)
- [4] **Se Yoon Lee** and Bani K. Mallick. (2021) “Bayesian Hierarchical modeling: application towards production results in the Eagle Ford Shale of South Texas,” *Sankhyā: The Indian Journal of Statistics, Series B* ; [[Github](#)]
- [5] **Se Yoon Lee**. (2021) “Gibbs sampler and coordinate ascent variational inference: a set-theoretical review,” *Communications in Statistics - Theory and Methods*
- [6] **Se Yoon Lee**, Alain Munafo, Pascal Girard, and Kosalaram Goteti. (2022) “Optimization of dose selection using multiple surrogates of toxicity as a continuous variable in Phase I cancer trial,” *Contemporary Clinical Trials* ; [[Github](#)]
- [7] **Se Yoon Lee**. (2022) “Bayesian Nonlinear Models for Repeated Measurement Data: An Overview, Implementation, and Applications,” *Mathematics*
- [8] **Se Yoon Lee**. (2022) “The Use of a Log-Normal Prior for the Student t-Distribution,” *Axioms*
- [9] Abhinav Prakash, **Se Yoon Lee**, Xin Liu, Lei Liu, Bani Mallick, Yu Ding. (2023) “A Bayesian hierarchical model to understand the effect of terrain on wind turbine power curves,” *IEEE Transactions on Sustainable Energy*
- [10] **Se Yoon Lee**. (2023) “A flexible dose-response modeling framework based on continuous toxicity outcomes in phase I cancer clinical trials,” *BMC Trials*
- [11] **Se Yoon Lee**. (2024) “Using Bayesian Statistics in Confirmatory Clinical Trials in the Regulatory Setting: A Tutorial Review,” *BMC Medical Research Methodology*
- [12] **Se Yoon Lee**. (2024) “Eliciting the discount parameter in a power prior method on the basis of the type I error consideration,” *Statistics in Biopharmaceutical Research*
- [13] **Se Yoon Lee**, Peng Zhao, Debdeep Pati, Bani K. Mallick. (2025) “Tail-adaptive Bayesian Shrinkage”, *Electronic Journal of Statistics*
- [14] **Se Yoon Lee**. (2025) “A note on the sample size formula for a win ratio endpoint’, *Statistics in Medicine*
- [15] **Se Yoon Lee**. (2025) “Power priors and type I error control: constrained borrowing of external control data, *Journal of Biopharmaceutical Statistics*
- [16] **Se Yoon Lee**. (2026) “Asymptotic validity of Schoenfeld’s sample size formula for the Cox proportional hazards model via the Wald test approach’, *Statistical Methods in Medical Research*

PROFESSIONAL ACTIVITIES

Editorial Board Member

- BMC Medical Research Methodology (Impact Factor: 3.4)

Invited Peer Reviewers

- Statistics in Medicine (Impact Factor: 2.373)
- Biostatistics (Impact Factor: 1.8)
- European Journal of Clinical Investigation (Impact Factor: 3.481)
- Artificial Intelligence Review (Impact Factor: 8.139)
- The Journal of Clinical Pharmacology (Impact Factor: 3.126)
- Frontiers in Public Health (Impact Factor: 3.709)
- The R Journal (Impact Factor: 3.984)
- Computational Geosciences (Impact Factor: 2.948)

- Journal of Applied Statistics (Impact Factor: 1.416)

HONORS AND AWARDS

Travel fund for an invited talk at Yonsei University	<i>Dec 2019</i>
Given to an invited speaker for the presentation	
Travel fund for an invited talk at University of Michigan	<i>Jun 2019</i>
Given to an invited speaker for an annual meeting about wind energy	
Travel fund for poster presenter for the Houston Geological Society	<i>Mar 2018</i>
Given to a poster presenter	
Anant Kshirsagar fellowship	<i>Jul 2018</i>
Given to graduate students who demonstrate excellence in making progress in research	
The 1st place prize in SETCASA poster session in 2018	<i>Apr 2018</i>
Given to only one winner for the poster session	
Graduate student travel award for JSM	<i>Jul 2017, 2018</i>
Given to graduate students who make a presentation at the conference	

INVITED TALKS

- “Important statistical issues in Bayesian medical device trials in regulatory settings” ; [\[Slide\]](#) *Sep 2023*
Organization: 2023 Orange County Biostatistics Symposium
- “Prediction of best-corrected visual acuity for wet age-related macular degeneration patients in HAWK and HARRIER studies via a Bayesian hierarchical linear model” *Aug 2020*
Organization: Novartis International AG, Internship Project Presentation
- “Estimation of COVID-19 spread curves integrating global data and borrowing information” *Jul 2020*
Organization: Mathophilia 2020, IQAC, Banwarilal Bhalotia College, Asansol, India
- “Bayesian Hierarchical Model: Application towards Wind Farm Data” *Jun 2020*
Organization: National Science Foundation, University of Connecticut, Mansfield, CT, U.S.A.
- “Continuous shrinkage prior revisited: a collapsing behavior and remedy” ; [\[Abstract\]](#) *Dec 2019*
Organization: Yonsei University, Department of Applied Statistics, Seoul, South Korea
- “Tutorial: understanding offshore wind energy data and spatial modeling” *Jun 2019*
Organization: National Science Foundation, University of Michigan, Ann Arbor, MI, U.S.A.

SOFTWARE

Software used in papers

- Bayesian Hierarchical Richards Model ; Written in R ; [\[Download\]](#) ; [\[Github\]](#)
- Spatial Weibull Model ; Written in R ; [\[Github\]](#)
- bayesestdft ; Written in R ; [\[Github\]](#)

SKILLS

Language	Fluent in English and Korean; Elementary in Chinese and Japanese
Computer Language	Proficient in R, Python, SQL, Microsoft Access, SAS, and NONMEM
Certificate	Cognigen NONMEM Workshop